4.1.4

If X is an object of $\mathcal{P}$ satisfying (ML), it follows from the expression of the associated graded recalled in (4.1.1) that, for every integer p, the projective system

$$\mathrm{gr}^p(X) = \big(\mathrm{gr}^p(X_n)\big)_{n\in\mathbb{N}}$$

is essentially constant.

Its projective limit, which exists without any hypothesis on the abelian category $\mathcal{C}$, will be called the p-th component of the strict associated graded of X, and will be denoted

$$\mathrm{grs}^p(X).$$

4.2. Projective systems adapted to J and the Artin-Rees lemma

We suppose given a commutative ring A, an ideal J of A, and an abelian A-category $\mathcal{C}$.

Definition 4.2.1. An object X of the category $\mathcal{P} = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ is said to be adapted to J if its filtration $(F^pX)_{p\in\mathbb{N}}$ as a projective system (4.1.1) is compatible with the J-adic filtration of X, i.e. if for every pair (p, q) of non-negative integers the inclusion below holds:

$$J^pF^q(X) \subset F^{p+q}(X).$$

Lemma 4.2.2. Let X be an object of the category $\mathcal{P}$. The following assertions are equivalent:

(i) X is adapted to J.

(ii) For every pair (n, p) of integers with $p \leq n+1$, one has

$$J^pF^{n-p+1}(X_n) = 0.$$

Proof. $(i) \Rightarrow (ii)$: then

$$J^pF^{n-p+1}(X_n) \subset F^{n+1}(X_n) = 0.$$

$(ii) \Rightarrow (i)$: if $n < p+q$, $F^{p+q}(X_n) = 0$ by definition. Moreover

$$J^pF^q(X_n)$$

is contained in $J^{n+1-q}F^q(X_n)$, and the latter is zero by hypothesis. If $n > p+q$, the image of $J^pF^q(X_n)$ in X_{p+q-1} under the transition morphism is contained in $J^pF^q(X_{p+q-1})$, which is zero by hypothesis. The inclusion

$$J^pF^q(X_n) \subset F^{p+q}(X_n)$$

therefore follows by definition of the filtration of a projective system.

In particular, it follows from (4.2.2) that every projective subsystem of a projective system adapted to J is adapted to J.

Examples. a) A J-adic projective system is adapted to J. It follows that every projective subsystem of a J-adic projective system is adapted to J.

b) If a projective system X satisfies condition (ML) and is adapted to J, then its projective system of universal images is also adapted to J.

4.2.3. The associated graded of a projective system adapted to J

Let X be a projective system adapted to J. Since the filtration of X is compatible with J, it is clear that the graded object

$$\big(\mathrm{gr}^p(X)\big)_{p\in\mathbb{N}}$$

is canonically endowed with the structure of a graded $\mathrm{gr}_J(A)$-module. In particular, by passage to the limit, one deduces that when X satisfies condition (ML), the graded object

$$\mathrm{grs}(X) = \big(\mathrm{grs}^p(X)\big)_{p\in\mathbb{N}}$$

is canonically endowed with the structure of a graded $\mathrm{gr}_J(A)$-module.

Let us briefly spell out how, when moreover X is strict, one defines the corresponding morphisms:

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q}).$$

Since the projective system X is strict, we know (4.1.1 and 4.1.2) that the following canonical morphism is an isomorphism:

$$\varphi : F^q(X_{p+q})/F^{q+1}(X_{p+q}) \longrightarrow F^q(X_q).$$

Moreover, the fact that the filtration of X is adapted to J gives a morphism

$$J^p \otimes_A F^q(X_{p+q}) \longrightarrow F^{p+q}(X_{p+q})$$

which factors, since

$$J^{p+1}F^q(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0, \qquad J^pF^{q+1}(X_{p+q}) \subset F^{p+q+1}(X_{p+q}) = 0,$$

through

$$J^p/J^{p+1} \otimes_A \big(F^q(X_{p+q})/F^{q+1}(X_{p+q})\big)$$

as a morphism

$$\beta : (J^p/J^{p+1}) \otimes_A \big(F^q(X_{p+q})/F^{q+1}(X_{p+q})\big) \longrightarrow F^{p+q}(X_{p+q}).$$

The morphism ω_{pq} is then given by the following expression:

$$\omega_{pq} = \beta \circ (\mathrm{id}_{J^p/J^{p+1}} \otimes_A \varphi)^{-1}.$$

Lemma 4.2.4. Let X be a J-adic projective system. The canonical morphism

$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X),$$

coming from the graded $\mathrm{gr}_J(A)$-module structure of $\mathrm{grs}(X)$, is an epimorphism.

Proof. This follows immediately from the description of the morphisms ω_{pq} given in (4.2.3).

Proposition 4.2.5. Characterization of strict AR-J-adic projective systems by means of their associated gradeds. Let X be a strict projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. Suppose that for every integer n, one has

$$J^{n+1}X_n = 0.$$

The following assertions are equivalent:

(i) X is AR-J-adic.

(ii) There exists an integer $r > 0$ such that, for every integer n, the following inclusion holds:

$$F^n(X_n) \subset J^{n-r}X_n.$$

(iii) There exists an integer r such that, for every integer m, the following canonical morphism is an isomorphism whenever $n \geq m + r$:

$$\mathrm{gr}^n(J^m X) \longrightarrow \mathrm{gr}^n(X).$$

In the statement of (ii), we agree to set $J^{n-r}X_n = X_n$ for $n < r$.

Proof. The equivalence of (ii) and (iii) is clear, since X and $J^m X$ are strict. Let us prove the equivalence of (i) and (ii). We know (3.2.3) that (i) is equivalent to the existence of an integer r such that, for every pair (m, n) of integers with $m \geq n + r$, the transition morphism below is an isomorphism:

$$X_{m+1}/J^{n+1}X_{m+1} \longrightarrow X_m/J^{n+1}X_m.$$

Consider the following exact commutative diagram:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & J^{n+1}X_{m+1} & \longrightarrow & J^{n+1}X_m & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & & & \\
0 & \longrightarrow & F^{m+1}(X_{m+1}) & \longrightarrow & X_{m+1} & \longrightarrow & X_m & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & & & \\
 & & X_{m+1}/J^{n+1}X_{m+1} & \longrightarrow & X_m/J^{n+1}X_m & \longrightarrow & 0 & & \\
 & & \downarrow & & \downarrow & & & & \\
 & & 0 & & 0. & & & &
\end{array}$$

It shows the existence of a canonical isomorphism between

$$\mathrm{Ker}\big(X_{m+1}/J^{n+1}X_{m+1} \longrightarrow X_m/J^{n+1}X_m\big)$$

and

$$F^{m+1}(X_{m+1})/\big(F^{m+1}(X_{m+1}) \cap J^{n+1}X_{m+1}\big).$$

In particular, for the transition morphism after reduction modulo J^{n+1} to be an isomorphism, it is necessary and sufficient that $F^{m+1}(X_{m+1})$ be contained in $J^{n+1}X_{m+1}$, which is precisely assertion (ii).

Theorem 4.2.6. Artin-Rees lemma. Let X be a projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. Suppose that:

(i) X satisfies (MLAR);

(ii) X is adapted to J;

(iii) $\mathrm{grs}(X)$ is a graded $(\mathrm{gr}_J(A), \mathcal{C})$-module of finite type, i.e. for every increasing sequence $(M_n)_{n\in\mathbb{N}}$ of graded sub-$\mathrm{gr}_J(A)$-modules of $\mathrm{grs}(X)$ whose union is $\mathrm{grs}(X)$, there exists an integer p such that

$$\mathrm{grs}(X) = M_p.$$

Then X is AR-J-adic.

Proof. Recall that if X' is the projective system of universal images of X, we have set

$$\mathrm{grs}(X) = \mathrm{gr}(X').$$

Moreover, X' is adapted to J, so it remains to show that X' is AR-J-adic. Suppose that the following lemma has been proved.

Lemma 4.2.7. Let X be a strict projective system adapted to J. For every integer s, the graded subobject $\big((M_s)_n\big)_{n\in\mathbb{N}}$ of $\mathrm{grs}(X)$ defined by the equalities

$$(M_s)_n = J^{n-s}X_n \cap F^n(X_n) \qquad (J^i = A \text{ if } i < 0),$$

is a graded $\mathrm{gr}_J(A)$-submodule of $\mathrm{grs}(X)$.

To prove the theorem, we may suppose X strict. Then $\mathrm{grs}(X)$ is the union of the increasing sequence of the M_s, hence is equal to one of them. This means that there exists an integer s_0 such that

$$F^n(X_n) \subset J^{n-s_0}X_n$$

for every integer n. We conclude by (4.2.5).

Proof of 4.2.7. Return to the notation of (4.2.3). We have to see that the morphism

$$\omega_{pq} : J^p/J^{p+1} \otimes_A F^q(X_q) \longrightarrow F^{p+q}(X_{p+q})$$

sends

$$J^p/J^{p+1} \otimes_A \big(J^{q-s}X_q \cap F^q(X_q)\big)$$

into

$$J^{p+q-s}X_{p+q} \cap F^{p+q}(X_{p+q}).$$

Now the canonical morphism

$$\rho : X_{p+q}/F^{q+1}(X_{p+q}) \longrightarrow X_q$$

is an isomorphism and sends

$$\big(J^{q-s}X_{p+q} + F^{q+1}(X_{p+q})\big)/F^{q+1}(X_{p+q})$$

onto $J^{q-s}X_q$. It follows that the inverse image under $\mathrm{id}(J^p/J^{p+1}) \otimes \varphi$ (4.2.3) of

$$J^p/J^{p+1} \otimes_A \big(J^{q-s}X_q \cap F^q(X_q)\big)$$

is
$$J^p/J^{p+1} \otimes_A \frac{(J^{q-s}X_{p+q} \cap F^q(X_{p+q})) + F^{q+1}(X_{p+q})}{F^{q+1}(X_{p+q})}.$$
It is now clear that the image under β (4.2.3) of this last object is contained in
$$J^{p+q-s}X_{p+q} \cap J^p F^q(X_{p+q}),$$
hence a fortiori in
$$J^{p+q-s}X_{p+q} \cap F^{p+q}(X_{p+q}).$$
This proves the lemma.

5. Noetherian J-adic and AR-J-adic projective systems

Throughout this paragraph, a noetherian commutative ring A, an ideal J of A, and an abelian A-category $\mathcal{C}$ are fixed.

5.1. Generalities

Definition 5.1.1. An object X of the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ is said to be noetherian J-adic (resp. artinian) if it is J-adic and if its components are noetherian (resp. artinian) objects of the category $\mathcal{C}$.

We shall denote by J-adn($\mathcal{C}$) (resp. J-adf($\mathcal{C}$)) the full subcategory of $\mathcal{P} = \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are noetherian (resp. artinian) J-adic projective systems. It is clear (3.1.3) that an extension of noetherian (resp. artinian) J-adic projective systems is of the same type.

Proposition 5.1.2. Let X be a J-adic object of $\mathcal{P}$. In order for X to be noetherian (resp. artinian) J-adic, it is necessary and sufficient that it be J-adic and that X_0 be noetherian (resp. artinian).

Proof. Indeed, let X be a J-adic projective system such that X_0 is noetherian (resp. artinian). Consider the canonical epimorphism (4.2.4):
$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X).$$
Since X_0 is noetherian (resp. artinian) and J^n is of finite type for every n, the existence of this epimorphism shows that for every integer n, the kernel of the canonical morphism
$$X_n \longrightarrow X_{n-1}$$
is noetherian (resp. artinian). It follows at once by induction on n that the X_n are noetherian (resp. artinian).

Definition 5.1.3. Let X be a projective system indexed by $\mathbb{N}$, with values in $\mathcal{C}$. It is said to be noetherian (resp. artinian) AR-J-adic if it is AR-J-adic and if its components are noetherian (resp. artinian) objects.

We shall denote by AR-J-adn($\mathcal{C}$) (resp. AR-J-adf($\mathcal{C}$)) the full subcategory of $\mathrm{Hom}_{\mathrm{AR}}(\mathbb{N}^\circ, \mathcal{C})$ (2.4) generated by the images of noetherian (resp. artinian) AR-J-adic projective systems.

It is then clear that the canonical functors

$$P^n_{\mathrm{AR}} : J\text{-adn}(\mathcal{C}) \longrightarrow \mathrm{AR}\text{-}J\text{-adn}(\mathcal{C}),$$

$$P^f_{\mathrm{AR}} : J\text{-adf}(\mathcal{C}) \longrightarrow \mathrm{AR}\text{-}J\text{-adf}(\mathcal{C}),$$

obtained by restriction of P_{AR} (2.4), are equivalences of categories.

Let us briefly recall the statement and proof of Hilbert's theorem.

Theorem 5.1.4. Hilbert's theorem. Let A be a commutative ring, $\mathcal{C}$ an abelian A-category, and M an object of finite type (resp. noetherian) of $\mathcal{C}$. For every A-algebra of finite type B, graded by $\mathbb{N}$, the graded $(B, \mathcal{C})$-module

$$B \otimes_A M$$

is of finite type (resp. noetherian).

N.B. If $B = \bigoplus_{n \geq 0} B_n$, then $B \otimes_A M$ denotes the graded object $\big(B_n \otimes_A M\big)_{n \in \mathbb{N}}$.

Proof. One reduces immediately to the case where B is the polynomial ring $A[T]$, endowed with its usual grading. The $(B, \mathcal{C})$-module $A[T] \otimes_A M = (M_n)_{n \in \mathbb{N}}$ is then obtained by putting $M_n = M$ for every n, with T sending M_n to M_{n+1} by the identity of M, for every integer n.

Let N be a graded sub-$(A[T], \mathcal{C})$-module of $A[T] \otimes_A M$. The components $(N_n)_{n \in \mathbb{N}}$ of N define an increasing sequence

$$N_0 \subset N_1 \subset \cdots \subset N_n \subset \cdots$$

of subobjects of M. Suppose now that M is noetherian (resp. of finite type), and let

$$N^1 \subset N^2 \subset \cdots \subset N^p \subset \cdots$$

be an increasing sequence of graded sub-$(A[T], \mathcal{C})$-modules of $A[T] \otimes_A M$, whose union, in the first hypothesis, when M is of finite type, is $A[T] \otimes_A M$.

Consider the following double-entry table:

$$\begin{array}{ccccccc}
N_0^1 & \subset & N_0^2 & \subset \cdots \subset & N_0^p & \subset \cdots \\
\cap & & \cap & & \cap & \\
N_1^1 & & N_1^2 & & N_1^p & \\
\vdots & & \vdots & & \vdots & \\
\cap & & \cap & & \cap & \\
N_n^1 & \subset & N_n^2 & \subset \cdots \subset & N_n^p & \subset \cdots .
\end{array}$$

It is clear by hypothesis on M that there exists a pair (p_0, n_0) such that $N^{p_0}_{n_0}$ is maximal (resp. equal to M). If p_1 is an integer $\geq p_0$ such that the $N^{p_1}_n$ $(n \leq n_0)$ are the largest elements of the sequences $(N^p_n)_{p \in \mathbb{N}}$, it is clear that $N^p = N^{p_1}$ as soon as $p \geq p_1$, which allows one to conclude in both cases.

Corollary 5.1.5. Suppose moreover that the category $\mathcal{C}$ has denumerable direct sums. Then for every object M of finite type (resp. noetherian) of $\mathcal{C}$, and every A-algebra of finite type B, the $(B, \mathcal{C})$-module $B \otimes_A M$ is of finite type (resp. noetherian).

Proof. One reduces to the case where $B = A[T]$. Let N be a sub-$(A[T], \mathcal{C})$-module of $A[T] \otimes_A M$, considered as an object of $\mathcal{C}$. We shall denote by $\mathrm{gr}(N)$ the associated graded of N for the filtration induced by the natural filtration of the graded object $A[T] \otimes_A M$: $\mathrm{gr}(N)$ is a graded sub-$(A[T], \mathcal{C})$-module of the graded $(A[T], \mathcal{C})$-module $A[T] \otimes_A M$.

Now let
$$N^0 \subset N^1 \subset \cdots \subset N^p \subset \cdots$$
be an increasing sequence of sub-$(A[T], \mathcal{C})$-modules of $A[T] \otimes_A M$, whose union is $A[T] \otimes_A M$ in the first hypothesis. In the first hypothesis (M of finite type), there exists an integer i such that $\mathrm{gr}(N^i) = A[T] \otimes_A M$, whence
$$N^i = A[T] \otimes_A M.$$
In the second hypothesis (M noetherian), the sequence of the $\mathrm{gr}(N^i)$ is stationary, hence the sequence of the N^i is stationary as well.

In what follows, Hilbert's theorem will be useful to us especially through the following proposition.

Proposition 5.1.6. Let A be a noetherian commutative ring, J an ideal of A, and $\mathcal{C}$ an abelian A-category. For every noetherian J-adic projective system X (5.1.1), the graded object $\mathrm{grs}(X)$ is a noetherian $(\mathrm{gr}_J(A), \mathcal{C})$-module.

Proof. Since $\mathrm{gr}_J(A)$ is an A-algebra of finite type, Hilbert's theorem (5.1.4) implies that
$$\mathrm{gr}_J(A) \otimes_A X_0$$
is a noetherian graded $(\mathrm{gr}_J(A), \mathcal{C})$-module. The assertion then follows from the existence of the epimorphism (4.2.4):
$$\mathrm{gr}_J(A) \otimes_A X_0 \longrightarrow \mathrm{grs}(X).$$

5.2. Structure of J-adn($\mathcal{C}$) and AR-J-adn($\mathcal{C}$)

In this section, we shall denote by $\mathcal{E}_{\mathcal{C}}$, or by $\mathcal{E}$ if there is no possible confusion, the full subcategory of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$ whose objects are noetherian AR-J-adic projective systems.

Proposition 5.2.1. The category $\mathcal{E}_{\mathcal{C}}$ is stable under kernels and cokernels in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$. In other words, $\mathcal{E}_{\mathcal{C}}$ is an abelian category and the inclusion functor
$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$$
is exact.

Proof. Let $u : X \to Y$ be a morphism of $\mathcal{E}_{\mathcal{C}}$. We shall successively see that the cokernel, the image, and the kernel of u are noetherian AR-J-adic. Of course, the proof that the image of u is noetherian AR-J-adic is only a technical intermediate step.

a) $\mathrm{Coker}(u)$ is noetherian AR-J-adic. Since X satisfies (MLAR) and Y is AR-J-adic, $\mathrm{Coker}(u)$ is AR-J-adic (3.2.4 (ii)).

b) $\mathrm{Im}(u)$ is noetherian AR-J-adic. From the exact sequence
$$0 \longrightarrow \mathrm{Im}(u) \longrightarrow Y \longrightarrow \mathrm{Coker}(u) \longrightarrow 0,$$
the assertion will follow from Lemma 5.2.1.1 below.

Lemma 5.2.1.1. Let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be an exact sequence in the category $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$. If M and N are noetherian AR-J-adic, then L is noetherian AR-J-adic.

Proof. It is plainly enough to see that it is AR-J-adic. One reduces, in the same way as in the proof of (3.2.4), to the case where M and N are strict, and then even noetherian J-adic. In this case, it follows from 3.1.3 (ii) that L is strict. The projective system L is strict, adapted to J, and its associated graded, as a graded subobject of $\mathrm{grs}(M)$, is noetherian: $\mathrm{grs}(M)$, a quotient of $\mathrm{gr}_J(A) \otimes_A M_0$, is noetherian (5.1.6). The assertion then follows from the Artin-Rees lemma (4.2.6).

c) $\mathrm{Ker}(u)$ is noetherian AR-J-adic. It is enough to apply (5.2.1.1) to the evident exact sequence below:

$$0 \longrightarrow \mathrm{Ker}(u) \longrightarrow X \longrightarrow \mathrm{Im}(u) \longrightarrow 0.$$

Proposition 5.2.2. The objects of the abelian category AR-J-adn$(\mathcal{C})$ are noetherian.

Proof. One reduces immediately to proving the following. Suppose we have a noetherian J-adic object X, a sequence of noetherian J-adic objects $(Y^n)_{n\in\mathbb{N}}$, and morphisms in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$

$$v_n : Y^n \longrightarrow X$$

such that

(i) the v_n become monomorphisms in AR-J-adn$(\mathcal{C})$;

(ii) for every integer n, v_n factors through v_{n+1} in AR-J-adn$(\mathcal{C})$,

then the v_n remain stationary, up to isomorphism, for sufficiently large n, in AR-J-adn$(\mathcal{C})$.

Now, to say that v_n factors through v_{n+1} in AR-J-adn$(\mathcal{C})$ means that there exists an integer r and a morphism

$$f : Y^n[r] \longrightarrow Y^{n+1}$$

such that, with a denoting the canonical morphism

$$Y^n[r] \longrightarrow Y^n,$$

the following equality holds:

$$v_n \circ a = v_{n+1} \circ f.$$

Since the morphism a is strict, it follows in particular that the image of v_n is contained in that of v_{n+1}. Replacing each Y^n by $\mathrm{Im}(v_n)$, we are reduced to seeing that every increasing sequence of strict projective subsystems of X is stationary. But if $(Z^n)_{n\in\mathbb{N}}$ is such a sequence, then, since $\mathrm{grs}(X)$ is noetherian, the increasing sequence of the $\mathrm{grs}(Z^n)$ is stationary, hence so is the sequence of the Z^n.

Summarizing:

Theorem 5.2.3. The categories J-adn$(\mathcal{C})$ and AR-J-adn$(\mathcal{C})$ are abelian and noetherian.

As for AR-J-adn$(\mathcal{C})$, this is what we have just proved. As for J-adn$(\mathcal{C})$, it is equivalent to it.

Proposition 5.2.4. Stability under extension. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. Suppose that:

(i) Z is AR-J-adic;

(ii) X is artinian AR-J-adic.

Then Y is AR-J-adic.

Proof. As in the proof of (3.2.4), one reduces to the case where Y and Z are strict and Z is J-adic. Then X is strict by (3.1.3 (ii)). There then exists an integer r such that the projective system $(X_{n+r}/J^{n+1}X_{n+r})_{n\in\mathbb{N}}$ is artinian J-adic. Since X is a quotient of it and is AR-isomorphic to it, we are reduced to proving the following lemma.

Lemma 5.2.4.1. Let

$$0 \longrightarrow T \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. If X is artinian J-adic, Z is J-adic, and T is AR-zero, then Y is AR-J-adic.

Proof. To prove the lemma, we first show that, for every integer n, the evident projective system

$$\left(Y_{n+k}/J^{n+1}Y_{n+k}\right)_{k\in\mathbb{N}}$$

is essentially constant. Let S_n^k denote the kernel of the evident morphism

$$X_{n+k}/J^{n+1}X_{n+k} \longrightarrow Y_{n+k}/J^{n+1}Y_{n+k}.$$

Consider the following evident exact commutative diagram:

$$\begin{array}{ccccccccc}
0 & \to & S_n^k & \to & X_{n+k}/J^{n+1}X_{n+k} & \to & Y_{n+k}/J^{n+1}Y_{n+k} & \to & Z_{n+k}/J^{n+1}Z_{n+k} \to 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
0 & \to & T_n & \to & X_n & \to & Y_n & \to & Z_n \qquad \to 0.
\end{array}$$

With n fixed, the objects S_n^k are identified with subobjects of T_n and form a decreasing sequence. Since T_n is artinian, there exists an integer k_0 such that $S_n^k = S_n^{k_0}$ for every integer $k \geq k_0$; it follows at once that the canonical morphisms

$$Y_{n+k}/J^{n+1}Y_{n+k} \longrightarrow Y_{n+k_0}/J^{n+1}Y_{n+k_0}$$

are isomorphisms.

This being said, denote by Y' the projective system defined by

$$Y'_n = \varprojlim_r Y_{n+r}/J^{n+1}Y_{n+r}$$

on objects, and in the evident way on morphisms. The projective system Y' is J-adic. Moreover, by passage to the limit one obtains an exact commutative diagram of the following type:

$$\begin{array}{ccccccccc} 0 \to & S & \to & X & \to & Y' & \to & Z & \to 0 \\ & \downarrow & & \downarrow & & \downarrow \mathrm{id} & & \downarrow \mathrm{id} & \\ 0 \to & T & \to & X & \to & Y & \to & Z & \to 0. \end{array}$$

Since T is AR-zero, the snake lemma applied to this diagram shows that the evident morphism $Y' \to Y$ is an AR-isomorphism, whence Y is AR-J-adic.

Corollary 5.2.5. Let

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

be an exact sequence in $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, with $J^{n+1}Y_n = 0$ for every n. Suppose that:

(i) X is artinian AR-J-adic (resp. artinian and noetherian);

(ii) Z is artinian AR-J-adic (resp. noetherian).

Then Y is artinian AR-J-adic (resp. noetherian).

5.3. Noetherian AR-J-adic projective systems and δ-functors

Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories, and let $F : \mathcal{C} \to \mathcal{D}$ be an additive functor. The functor F extends in the clear way to a functor, again denoted F,

$$F : \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{D}),$$

which is additive and "commutes with translations".

Proposition 5.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be two abelian A-categories and

$$T = (T^i)_{i \in \mathbb{N}}$$

an exact A-linear δ-functor from $\mathcal{C}$ to $\mathcal{D}$. Suppose that:

(i) For every noetherian object M of $\mathcal{C}$, annihilated by a power of J, and every integer n, $T^n(M)$ is noetherian.

(ii) For every graded A-algebra B of finite type, annihilated by J, and every noetherian graded $(B, \mathcal{C})$-module N, the graded $(B, \mathcal{D})$-modules $T^n(N)$, $n \in \mathbb{N}$, are noetherian.

Then, for every noetherian AR-J-adic object X of $\mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C})$, the $T^n(X)$ are noetherian AR-J-adic.

Proof. Since the functors T^n commute with translations, it is enough to prove the assertion when X is noetherian J-adic. In that case, since $\mathrm{grs}(X)$ is a noetherian graded $\mathrm{gr}_J(A)$-module (5.1.6), it follows from hypothesis (ii) and from a variant of a corollary of Shih's theorem (cf. Appendix A.3) that for every integer n the following assertions hold:

$$\text{(a)} \quad T^n(X) \text{ satisfies (MLAR)},$$

(b) grs $T^n(X)$ is a noetherian graded $(\mathrm{gr}_J(A), \mathcal{D})$-module.

The Artin-Rees lemma (4.2.6) allows us to conclude.

Corollary 5.3.2. The δ-functor

$$T = (T^i)_{i\in\mathbb{N}} : \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{N}^\circ, \mathcal{D})$$

induces an exact A-linear δ-functor

$$\mathcal{E}_{\mathcal{C}} \longrightarrow \mathcal{E}_{\mathcal{D}},$$

commuting with translations, and hence, by passage to the quotient, a new exact A-linear δ-functor, again denoted $(T^i)_{i\in\mathbb{N}}$,

$$\text{AR-}J\text{-adn}(\mathcal{C}) \longrightarrow \text{AR-}J\text{-adn}(\mathcal{D}).$$

Appendix: Shih's theorem

The purpose of the present appendix is to give a proof of Shih's theorem in the form used in the proof of (5.3.1). This form being more general than the one appearing in EGA 0_{III}, a direct proof, adapted from that of loc. cit., has proved necessary.

Terminology. Let r_0 be an integer ≥ 1. The datum, for every integer $r \geq r_0$, of a bigraded object

$$E_r = \left(E_r^{pq}\right)_{p\in\mathbb{Z},\ q\in\mathbb{Z}},$$

endowed with a differential d_r of bidegree $(r, 1-r)$, and, for every r, of an isomorphism

$$H(E_r) \xrightarrow{\sim} E_{r+1},$$

will be called below a spectral filter. The notion of a morphism of spectral filters is defined in the evident way.

A.1. Recollections on the spectral sequences associated with δ-functors

Let $\mathcal{C}$ be an abelian category and X an object of $\mathcal{C}$, endowed with a decreasing filtration $(F^pX)_{p\in\mathbb{Z}}$. For every exact δ-functor T from $\mathcal{C}$ to an abelian category $\mathcal{D}$, the method of "exact couples" makes it possible to describe a spectral sequence

$$E(X): \quad E_1^{pq} = T^{p+q}(F^pX/F^{p+1}X) \Rightarrow E^n = T^n(X),$$

for which the filtration of E^n is defined as follows:

$$F^p(E^n) = \mathrm{Im}(T^n(F^pX) \longrightarrow T^n(X)) = \mathrm{Ker}(T^n(X) \longrightarrow T^n(X/F^pX)).$$

Moreover, if the filtration of X is finite, the spectral sequence so obtained is weakly convergent (EGA 0_{III} 11.1.3).

Functoriality. If $f : X \to Y$ is a morphism of filtered objects of $\mathcal{C}$, one constructs functorially a morphism of spectral sequences

$$E(f) : E(X) \longrightarrow E(Y),$$

coinciding with $T^{p+q}(\mathrm{gr}^p f)$ (resp. $T^n(f)$) on $E_1^{pq}(X)$ (resp. $E^n(X)$). In particular, if $u : X \to X$ is an endomorphism of X such that

$$u(F^p X) \subset F^{p+h}(X)$$

for an integer h independent of p, then $E(u)$ is an endomorphism of bidegree $(h, -h)$ of the spectral filter associated with $E(X)$, commuting with the differentials.

A.2. The case of projective systems

Let now $X = (X_n)_{n\in\mathbb{Z}}$ be a projective system of objects of an abelian category $\mathcal{C}$, such that

$$X_i = 0 \qquad (i < 0).$$

If T denotes an exact δ-functor from $\mathcal{C}$ to an abelian category $\mathcal{D}$, T extends naturally to an exact δ-functor, again denoted T,

$$\mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{C}) \longrightarrow \mathrm{Hom}(\mathbb{Z}^\circ, \mathcal{D}).$$

Thus, endowing X with its natural filtration as a projective system (EGA 0_{III} 13.4.1), one obtains a spectral sequence

$$E(X) = E(X_n)_{n\in\mathbb{Z}},$$

where the $E(X_n)$ denote the spectral sequences associated with T and the X_n endowed with the induced filtration.

Properties of $E(X)$. a) For every integer n, the filtration of the abutment $E^n(X)$ in the spectral sequence $E(X)$ is identical with its natural filtration as a projective system when X is strict.

Indeed, denoting by the letter F the first filtration and by the letter Φ the second, one has for every integer α:

$$\begin{aligned} {}^pF\,E^n(X)_\alpha &= \mathrm{Ker}\big(T^n(X_\alpha) \longrightarrow T^n(X_\alpha/F^pX_\alpha)\big) \\ &= \begin{cases} 0, & \alpha \le p-1, \\ \mathrm{Ker}\big(T^n(X_\alpha) \longrightarrow T^n(X_{p-1})\big), & \alpha \ge p-1. \end{cases} \end{aligned}$$

In both cases one recognizes the expression of $\Phi^p E^n(X)$.

This shows in particular that, for $\beta \ge \alpha \ge p$, the transition morphism

$$E_\infty^{pq}(X_\beta) \longrightarrow E_\infty^{pq}(X_\alpha)$$

which is identified with

$$\mathrm{gr}_T^p\, T^{p+q}(X_\beta) \longrightarrow \mathrm{gr}_T^p\, T^{p+q}(X_\alpha),$$

is a monomorphism, by EGA 0_{III} 13.4.1.3.

b) Fixing $\alpha \in \mathbb{Z}$, it is clear that, since $F^i(X_\alpha) = 0$ for $i \ge \alpha + 1$, one has

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \quad \text{for } r > p,$$

$$Z_r^{pq}(X_\alpha) = Z_{r+1}^{pq}(X_\alpha) \quad \text{for } r > \alpha - p.$$

c) When X satisfies the Mittag-Leffler condition, the various projective systems $\mathrm{gr}^p(X)$ are essentially constant (EGA 0_{III} 13.4.3). It follows at once that, for $r < +\infty$, the projective systems $E_r^{pq}(X)$ are essentially constant. We shall denote by

$$\mathcal{E}_r^{pq}(X)$$

their projective limits. Since the differential operators pass to the limit, one thus obtains a spectral filter $\mathcal{E}_r^{pq}(X)$, $p, q \in \mathbb{Z}$, $r \geq 1$. The objects

$$\mathcal{Z}_r^{pq}(X), \quad \mathcal{B}_r^{pq}(X) \quad (r < +\infty)$$

associated with this spectral filter are precisely the limits of the essentially constant projective systems $Z_r^{pq}(X)$ and $B_r^{pq}(X)$.

It also follows from part b) that

$$B_\infty^{pq}(X) = B_{p+1}^{pq}(X),$$

which allows one to put

$$\mathcal{B}_\infty^{pq}(X) = \mathcal{B}_{p+1}^{pq}(X).$$

Still supposing that X satisfies (ML), since, for every α,

$$B_r^{pq}(X_\alpha) = B_{r+1}^{pq}(X_\alpha) \qquad (r > p),$$

one has a sequence of canonical monomorphisms

$$\cdots \subset \mathcal{E}_r^{pq}(X) \subset \cdots \subset \mathcal{E}_{p+2}^{pq}(X) \subset \mathcal{E}_{p+1}^{pq}(X). \tag{S}$$

Suppose from now on that X is strict. The following expressions for $Z_r^{pq}(X_n)$ and $B_r^{pq}(X_n)$:

$$Z_r^{pq}(X_n) = \mathrm{Im}\Big[T^{p+q-1}(F^{p-r+1}X_n/F^pX_n) \xrightarrow{\partial} T^{p+q}(F^pX_n/F^{p+1}X_n)\Big],$$

$$B_r^{pq}(X_n) = \mathrm{Im}\Big[T^{p+q}(F^pX_n/F^{p+r}X_n) \longrightarrow T^{p+q}(F^pX_n/F^{p+1}X_n)\Big],$$

show that

$$\begin{cases} Z_r^{pq}(X_m) \xrightarrow{\sim} Z_r^{pq}(X_n) & \text{for } m \geq n \geq p, \\ B_r^{pq}(X_m) \xrightarrow{\sim} B_r^{pq}(X_n) & \text{for } m \geq n \geq p+r-1, \end{cases} \tag{d_1}$$

where the canonical morphisms are isomorphisms. In particular, the canonical morphism

$$\mathcal{E}_r^{pq}(X) \longrightarrow E_r^{pq}(X_{p+r-1})$$

is an isomorphism. Moreover, it follows from b) that

$$\begin{cases} Z_\infty^{pq}(X_{p+r-1}) = Z_r^{pq}(X_{p+r-1}) & (r \geq 1), \\ B_\infty^{pq}(X_{p+r-1}) = B_r^{pq}(X_{p+r-1}) & (r > p). \end{cases} \tag{d_2}$$

This makes it possible to define, for every r, a canonical epimorphism

$$\theta_r^{pq} : \mathcal{E}_r^{pq}(X) \longrightarrow E_\infty^{pq}(X_{p+r-1})$$

which is an isomorphism when $r > p$. In addition, if a is an integer such that

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \qquad (r \geq a),$$

for example $a = p+1$ suffices, so that

$$\cdots \hookrightarrow \mathcal{E}_r^{pq}(X) \hookrightarrow \cdots \hookrightarrow \mathcal{E}_{a+1}^{pq}(X) \hookrightarrow \mathcal{E}_a^{pq}(X), \tag{S bis}$$

then, for $s \geq r \geq a$, the diagrams

$$\begin{array}{ccc} \mathcal{E}_s^{pq}(X) & \xrightarrow{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\ \downarrow & & \downarrow \text{can} \\ \mathcal{E}_r^{pq}(X) & \xrightarrow{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1}) \end{array} \tag{d_3}$$

are commutative.

A.3. A variant of Shih's theorem

Suppose given a ring S filtered by ideals $F^i(S)$, $i \geq 0$, acting on X compatibly with its filtration as a projective system.

It follows from the generalities of A.1 that, for every integer i, $\mathrm{gr}^i(S)$ acts on the bigraded objects $E_r(X)$ by morphisms of bidegree $(i, -i)$. In particular, for every integer n,

$$E_r^{(n)}(X) = \left(E_r^{p,q}(X)\right)_{p+q=n}$$

is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module. The same holds by passage to the limit for

$$\mathcal{E}_r^{(n)}(X) = \left(\mathcal{E}_r^{p,q}(X)\right)_{p+q=n}.$$

The ring S, acting on X compatibly with its filtration as a projective system, likewise acts on $T^n(X)$ for every integer n. Consequently, the strict associated graded $\mathrm{grs}(X)$ is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{C})$-module, and similarly, when $T^n(X)$ satisfies (ML), the strict associated graded $\mathrm{grs}\, T^n(X)$ is canonically endowed with the structure of a graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module.

Theorem A.3. Suppose that, for an integer n, the graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules

$$\mathcal{E}_1^{(n)}(X) = T^n(\mathrm{grs}^\bullet X) \quad \text{and} \quad \mathcal{E}_1^{(n-1)}(X) = T^{n-1}(\mathrm{grs}^\bullet X)$$

are noetherian. Then:

(i) $T^n(X)$ satisfies (MLAR);

(ii) $\mathrm{grs}\, T^n(X)$ is a noetherian $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module.

Proof. i) For every pair (m, r) of integers,

$$\mathcal{B}_r^{(m)}(X) = \left(\mathcal{B}_r^{p,m-p}(X)\right)_{p\in\mathbb{N}}$$

is a graded sub-$(\mathrm{gr}^\bullet(S), \mathcal{D})$-module of $\mathcal{E}_1^{(m)}(X)$. In particular, for every integer m, the $\mathcal{B}_r^{(m)}(X)$ form an increasing sequence, indexed by r, of graded sub-$(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules of $\mathcal{E}_1^{(m)}(X)$. Using the fact that $\mathcal{E}_1^{(n)}(X)$ and $\mathcal{E}_1^{(n-1)}(X)$ are noetherian graded $(\mathrm{gr}^\bullet(S), \mathcal{D})$-modules, one deduces that the sequences

$$\left[\mathcal{B}_r^{(n)}(X)\right]_{r\in\mathbb{N}} \qquad \text{and} \qquad \left[\mathcal{B}_r^{(n-1)}(X)\right]_{r\in\mathbb{N}}$$

are stationary.

Thus there exists an integer $a \geq 1$ such that, for every (p,q), with $p+q=n$ or $n-1$,

$$\mathcal{B}_r^{pq}(X) = \mathcal{B}_a^{pq}(X) \qquad (r \geq a). \tag{A.3.1}$$

Consequently, for $r \geq a+1$, the arrows d_r with source and target $\mathcal{E}_r^{pq}(X)$, $p+q=n$, are zero. In other words, the monomorphisms

$$\mathcal{E}_s^{pq}(X) \hookrightarrow \mathcal{E}_r^{pq}(X) \qquad (s \geq r \geq a+1)$$

deduced from (A.3.1) with $p+q=n$ are in fact isomorphisms. For $s \geq r \geq a+1$, the commutativity of diagram (d_3), with $p+q=n$,

$$\begin{array}{ccc}
\mathcal{E}_s^{pq}(X) & \xrightarrow{\theta_s^{pq}} & E_\infty^{pq}(X_{p+s-1}) \\
\downarrow\wr & & \downarrow \\
\mathcal{E}_r^{pq}(X) & \xrightarrow{\theta_r^{pq}} & E_\infty^{pq}(X_{p+r-1})
\end{array}$$

shows that the transition morphism

$$E_\infty^{pq}(X_{p+s-1}) \longrightarrow E_\infty^{pq}(X_{p+r-1}),$$

which is in any case a monomorphism (cf. a)), is an epimorphism, hence an isomorphism. In other words, the projective system

$$r \longmapsto \left[\bigoplus_p \mathrm{gr}^p\, T^n(X_{p+r})\right]_{r\geq 1}$$

is constant from $r=a$ on, and therefore $T^n(X)$ satisfies (MLAR), by (4.1.3).

ii) Denoting by

$$\mathcal{E}_\infty^{(n)}(X), \quad \mathcal{Z}_\infty^{(n)}(X), \quad \mathcal{B}_\infty^{(n)}(X)$$

the "limits" of the various projective systems indexed by r considered above, we see that

$$\mathcal{E}_\infty^{(n)}(X) = \mathcal{Z}_\infty^{(n)}(X)/\mathcal{B}_\infty^{(n)}(X)$$

is a noetherian $(\mathrm{gr}^\bullet(S), \mathcal{D})$-module, since $\mathcal{Z}_\infty^{(n)}(X)$ is contained in $\mathcal{E}_1^{(n)}(X)$. Moreover, it follows from A.2 a) that

$$\mathcal{E}_\infty^{(n)}(X) = \mathrm{grs}\; T^n(X).$$